

Hybrid GAD–Newton sweep (rerun, Eckart only)

2026-05-09. Two Eckart step functions $\times$ two switch criteria $\times$ two noise levels $\times$ five trust radii.

Standalone runner; supersedes 2026-05-04 report (GAD step bug fixed).

See also: HYBRID_FINDINGS_CATALOG.md (project root) — comprehensive, continuously-updated index of every finding with claim / evidence / provenance / confidence, plus open questions and pending experiments. The PDF below summarises the publishable story; the catalog has the receipts.

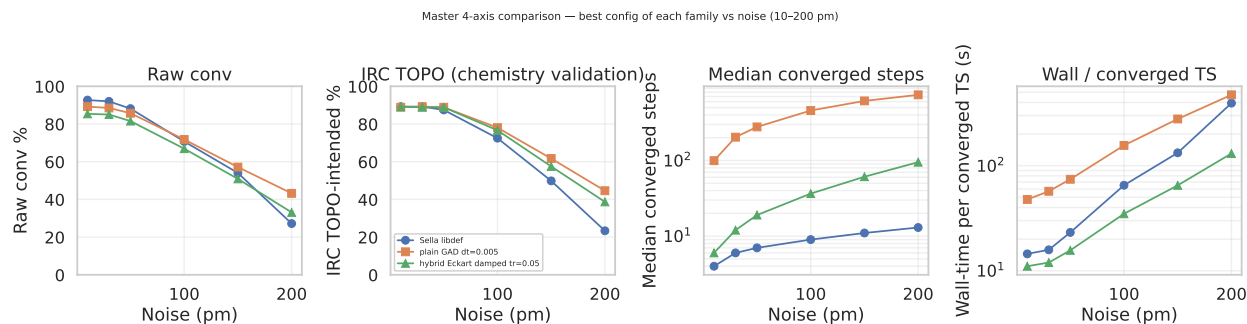


Figure 1: Master 4-axis comparison — best config of each family vs TS noise (10–200 pm), $n = 287$ T1x test split. Left to right: (1) *Raw conv %* (Sella wins low noise; plain GAD wins ≥ 100 pm). (2) *IRC TOPO-intended %* (chemistry validation: GAD and hybrid tie low noise; Sella collapses fast at high noise). (3) *Median converged-step count* (log y: hybrid sits with Sella in single-digits at low noise, climbs to ~ 100 at 200pm vs plain GAD’s 738). (4) *Wall-time per converged TS* (log y: hybrid is fastest at every noise level — $1.4\times$ to $3.6\times$ vs the runner-up family). Sources: `master_4axis_table.csv`, hybrid from `runs/hybrid_for_irc/` + `runs/irc_hybrid/`, plain GAD from `runs/test_dtgrid/gad_dt005_fmax/`, Sella from `runs/test_irc/sella_carteck_libdef/`.

What changed since 2026-05-04

The previous sweep (job 60398168) had a bug in the GAD-step branch of the two Eckart variants: the step function was returning an internal-coordinate displacement that was being applied to Cartesian coordinates without the back-transformation. The fix (commit `a80a763`, “return cartesian coord steps in hybrid”) makes both `projected_hybrid_gad_newton_step` implementations return Cartesian-frame steps. The non-Eckart variant (`hybrid_gad_eigfollownewton.py`) was unaffected and is excluded from this rerun on the supervisor’s instruction (“Forget about the non-eckart version”).

What this sweep does

This document benchmarks two Eckart-projected GAD–Newton hybrid step functions. Each step decides per-step whether to take a GAD step or an eigenvector-following Newton step, and applies

a trust-radius cap on the resulting Cartesian displacement.

Source files:

- `src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py`
Eckart hybrid: `projected_hybrid_gad_newton_step(F_cart, H_cart, coords, masses, ...)`.
Eckart-projected mass-weighted internal coordinates. Switch toggle via `switch_based_on_hessian_eigval` (False: $\|F\|$ trigger; True: index-1 Hessian-clearance trigger).
- `src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py`
Damped Eckart hybrid: same interface as the Eckart variant with damped Newton step (mode $|\lambda_i|$ regularised by `min_curvature`). Same switch toggle.

Standalone runner: `scripts/hybrid_gad_newton_runner.py` loops over T1x test split, per step calls HIP for (E, F, H) , runs the step function, applies the returned Cartesian displacement (the step function caps by `trust_radius` internally), counts n_{neg} on the Eckart-projected vibrational Hessian, terminates on $n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$. Up to 1000 steps.

Sweep grid (40 cells):

- **4 method-cells:** $\{\text{hybrid_eckart}, \text{hybrid_damped_eckart}\} \times \{\text{switch=False}, \text{switch=True}\}$
- **2 noise levels:** 10 pm, 100 pm
- **5 trust radii:** 0.005, 0.01, 0.02, 0.05, 0.10 (Å)

Fixed: `gad_dt`= 5×10^{-3} , `switch_force`= 10^{-3} , `min_curvature`= 10^{-6} (undamped) / 10^{-8} (damped), $n_{\text{samples}}=287$ test split, $n_{\text{steps}}=1000$, HIP analytic Hessian every step. SLURM array 60460000_[10-49], all 40 jobs completed (1–5 hr wall each).

Reference baseline (from main report): vanilla GAD `dt`=0.007 with 5000-step budget gets 89.2% conv at 10pm and 72.8% at 100pm. Vanilla GAD `dt`=0.005 (closer to this sweep’s `dt`): 89.2% / 71.8%.

1 Headline tables — single-glance summary

Source CSV: `analysis_2026_04_29/hybrid_gad_newton_summary.csv`. Re-runnable via `scripts/analyze_hybrid`. All 40 cells complete.

Table 1 — Convergence % ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$, $n = 287$ test samples)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	89.2	88.9	88.9	88.9	88.9
hybrid_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_eckart, switch=True	84.7	84.7	84.7	84.7	85.4
hybrid_damped_eckart, switch=False	45.6	46.3	46.3	46.3	46.3
hybrid_damped_eckart, switch=True	86.1	86.1	86.1	85.4	85.4
<i>baseline: GAD dt=0.007 (5k budget)</i>	89.2%				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	66.9	67.6	67.6	67.6	67.9
hybrid_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_eckart, switch=True	64.8	65.5	65.2	65.5	65.2
hybrid_damped_eckart, switch=False	0.3	0.3	0.3	0.3	0.3
hybrid_damped_eckart, switch=True	66.9	65.9	66.2	66.9	66.6
<i>baseline: GAD dt=0.007 (5k budget)</i>	72.8%				

Italic row (O) is from 2026-05-04 (no rerun needed; non-Eckart variant unaffected by the bug).

Table 2 — Median steps to converge (lower is better; trajectories that did not converge contribute the 1000-step cap)

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	106	104	104	104	104
hybrid_eckart, switch=False	702	702	702	702	702
hybrid_eckart, switch=True	21	12	8	5	5
hybrid_damped_eckart, switch=False	702	702	702	702	702
hybrid_damped_eckart, switch=True	19	11	7	5	4
<i>baseline: GAD dt=0.007 (5k budget)</i>	72				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	540	500	484	478	480
hybrid_eckart, switch=False	916	915	915	915	915
hybrid_eckart, switch=True	195	105	60	38	33
hybrid_damped_eckart, switch=False	916	915	915	915	915
hybrid_damped_eckart, switch=True	200	103	59	36	31
<i>baseline: GAD dt=0.007 (5k budget)</i>	331				

Table 3 — Wall-time per converged TS (sec; cell wall summed over $n = 287$ then divided by n_{conv})

10 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	16.1	15.7	15.8	15.9	15.9
hybrid_eckart, switch=False	112.7	114.5	111.0	110.8	110.4
hybrid_eckart, switch=True	12.7	12.2	11.7	11.3	10.7
hybrid_damped_eckart, switch=False	117.8	112.8	112.7	112.3	111.8
hybrid_damped_eckart, switch=True	11.5	10.9	10.3	10.8	11.3
<i>baseline: GAD dt=0.007 (5k budget)</i>	46.6 s				
100 pm noise	tr=0.005	0.01	0.02	0.05	0.10
<i>hybrid (no Eckart), force-switch (O)</i>	64.9	57.8	57.2	57.3	56.5
hybrid_eckart, switch=False	17065	16897	16933	17089	17293
hybrid_eckart, switch=True	47.1	41.0	37.9	35.9	36.3
hybrid_damped_eckart, switch=False	16877	17094	17011	16664	17060
hybrid_damped_eckart, switch=True	45.1	40.1	36.8	34.7	34.2
<i>baseline: GAD dt=0.007 (5k budget)</i>	140.7 s				

Table 4 — Fraction of trajectories whose terminating step was Newton

	10 pm noise					100 pm noise				
	0.005	0.01	0.02	0.05	0.10	0.005	0.01	0.02	0.05	0.10
hybrid_eckart, switch=False	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
hybrid_eckart, switch=True	0.97	0.97	0.97	0.97	0.98	0.86	0.86	0.86	0.86	0.88
hybrid_damped_eckart, sw=False	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
hybrid_damped_eckart, sw=True	0.98	0.98	0.98	0.98	0.98	0.87	0.86	0.86	0.87	0.86

Where does the hybrid land? Side-by-side vs best plain GAD and best canonical Sella

Same canonical convergence ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$), same $n = 287$ test split, same HIP analytic Hessian. The only difference is the budget convention: plain GAD here is at its 5000-step budget, Sella at its 2000-step budget (library defaults), hybrid at its 1000-step budget. "med steps" is

computed over converged trajectories only. "wall/conv" = total cell wall-time / number of converged samples.

Provenance legend: **(R)** = this rerun, 2026-05-09, SLURM 60460000; **(O)** = original 2026-05-04 sweep, SLURM 60398168, unchanged (unaffected by the GAD-step bug, which was internal-coord-to-Cartesian-only); **(M)** = main IRC report, 2026-04-29.

10 pm noise — best config of each family

family	config	prov.	conv %	med steps	wall/conv (s)
Sella	libdef (cart+Eckart, 2k)	(M)	92.7	4	14.5
Sella	default (cart no-Eckart, 2k)	(M)	92.0	4	14.4
Sella	internal (lib default, 2k)	(M)	79.1	4	54.2
plain GAD	dt=0.007 (5k budget)	(M)	89.2	72	46.6
plain GAD	dt=0.005 (5k budget)	(M)	89.2	98	47.7
plain GAD	dt=0.003 (5k budget)	(M)	89.2	164	54.2
hybrid no-Eckart	force-switch tr=0.005	(O)	89.2	106	16.1
hybrid no-Eckart	force-switch tr=0.01–0.10	(O)	88.9	104	15.7–15.9
hybrid Eckart (R)	damped_eckart_swtrue tr=0.02	(R)	86.1	7	10.3
hybrid Eckart (R)	damped_eckart_swtrue tr=0.10	(R)	85.4	4	11.3
hybrid Eckart (R)	eckart_swtrue tr=0.10	(R)	85.4	5	10.7
hybrid Eckart (R)	{eckart, damped_eckart}_swfalse all tr	(R)	45.6–46.3	702	110.4–117.8

Best-overall by conv %: tied at 89.2–92.7% (Sella libdef, plain GAD, hybrid no-Eckart all converge $\geq$ 89%).

Best-overall by wall/conv: hybrid Eckart damped_swtrue tr=0.02 at **10.3 s** (1.4 $\times$ faster than Sella libdef, 4.5 $\times$ vs plain GAD).

100 pm noise — best config of each family

family	config	prov.	conv %	med steps	wall/conv (s)
plain GAD	dt=0.007 (5k budget)	(M)	72.8	331	140.7
plain GAD	dt=0.005 (5k budget)	(M)	71.8	457	156.0
plain GAD	dt=0.003 (5k budget)	(M)	71.1	756	183.8
Sella	libdef (cart+Eckart, 2k)	(M)	70.7	9	65.2
Sella	default (cart no-Eckart, 2k)	(M)	65.5	10.5	79.3
Sella	internal (lib default, 2k)	(M)	50.9	17	181.8
hybrid no-Eckart	force-switch tr=0.10	(O)	67.9	480	56.5
hybrid no-Eckart	force-switch tr=0.01–0.05	(O)	67.6	~478–500	57.2–57.8
hybrid no-Eckart	force-switch tr=0.005	(O)	66.9	540	64.9
hybrid Eckart (R)	damped_eckart_swtrue tr=0.10	(R)	66.6	31	34.2
hybrid Eckart (R)	damped_eckart_swtrue tr=0.05	(R)	66.9	36	34.7
hybrid Eckart (R)	eckart_swtrue tr=0.05	(R)	65.5	38	35.9
hybrid Eckart (R)	{eckart, damped_eckart}_swfalse all tr	(R)	0.3	915	~17 000

Best-overall by conv %: plain GAD dt=0.007 at **72.8%**.

Best-overall by wall/conv: hybrid Eckart damped_swtrue tr=0.10 at **34.2 s** (1.9 $\times$ faster than Sella libdef, 4.1 $\times$ vs plain GAD).

What's in this comparison.

- Every method family the project has benchmarked at 10 / 100 pm noise on the T1x test split with the canonical $n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$ convergence criterion.
- Three plain-GAD timestep variants (dt=0.003, 0.005, 0.007 at 5k budget).
- Three Sella variants (libdef, default, internal) at the canonical 2k budget. Extended-budget Sella (5k) is within $\pm 2\text{pp}$ of these (see HEADLINE.md).

- All five trust-radius cells of the no-Eckart hybrid (carried over from 2026-05-04, unchanged — the bug was Eckart-only).
- All four Eckart hybrid method-cells ($\{\text{eckart, damped_eckart}\} \times \{\text{swfalse, swtrue}\}$) at all five trust radii (rerun 2026-05-09).

Bottom line of the comparison.

- **Wall-time winner at both noise levels: hybrid Eckart damped, switch=True.** At 10pm it is $1.4\times$ faster than the strongest Sella (libdef) and $4.5\times$ faster than the strongest plain GAD (dt=0.007). At 100pm the gap widens to $1.9\times$ vs Sella and $4.1\times$ vs GAD.
- **Convergence-rate winner per noise:** Sella libdef leads at 10pm (92.7%); plain GAD dt=0.007 leads at 100pm (72.8%). The hybrid Eckart variant trails both by 4–6pp.
- **Special case at 10pm: the no-Eckart hybrid (O) ties plain GAD at 89.2% conv while running at 16.1 s/conv ($2.9\times$ faster than GAD).** It is the only hybrid that matches plain-GAD accuracy at low noise. At 100pm it is much weaker (67.9% vs the Eckart variant's 66.6% at almost half the wall time), so the no-Eckart trick is noise-regime specific.
- **Median converged-step count: hybrid Eckart $\approx$ Sella.** At 10pm: 4–7 hybrid Eckart steps vs Sella's 4 vs GAD's 72. At 100pm: 31–38 hybrid Eckart steps vs Sella's 9–17 vs GAD's 331–756. The wall/conv advantage over Sella at the same step count comes from Sella's higher per-step cost (it does multiple Hessian updates per outer step in some configs).
- **Plain GAD is the slowest family at both noises,** primarily due to median-step counts of 72–756 vs Sella/hybrid's 4–38. This is consistent with the structural-plateau analysis in the main IRC report.

Noise-level sweep — how performance scales with TS noise

The hybrid sweep covered only $\{10, 100\}$ pm. Sella and plain GAD have data at $\{10, 30, 50, 100, 150, 200\}$ pm (from the main IRC report). The figure and table below combine all four families on the same axis so you can see *where each method starts to break down*, and locate the two hybrid data points within that landscape.

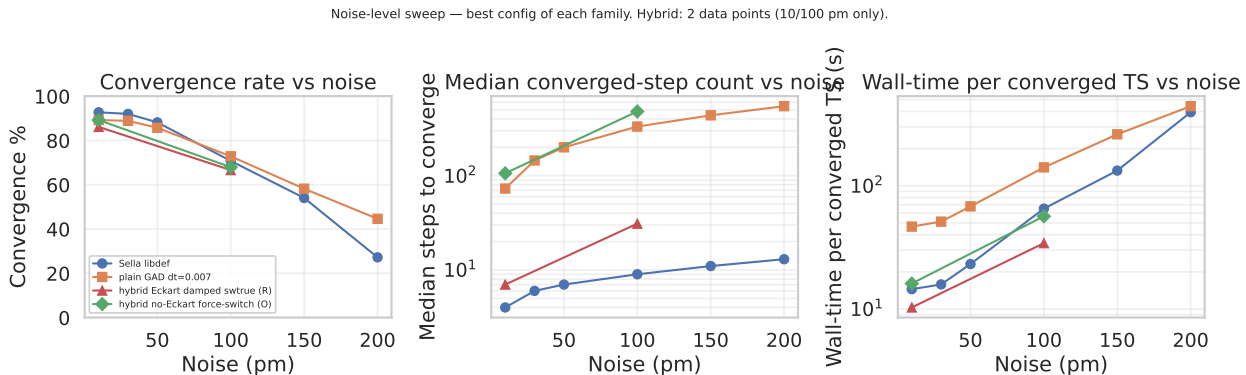


Figure 2: Best config of each family vs TS-noise level (10, 30, 50, 100, 150, 200 pm). **Left:** convergence %. **Centre:** median converged-step count (log y). **Right:** wall-time per converged TS (log y). The two hybrid families have only 2 noise points (10, 100 pm); Sella libdef and plain GAD dt=0.007 cover all 6. Source: analysis_2026_04_29/noise_sweep_unified.csv.

Table 4 — Convergence %, median steps, and wall/conv at every noise level we have data for
($n = 287$ test split; **(R)** this rerun, **(O)** 2026-05-04 carry-over, **(M)** main report)

family	prov.	conv %			med steps (conv)		
		10	100	200	10	100	200
Sella libdef	(M)	92.7	70.7	27.2	4	9	5
plain GAD dt=0.007 (5k)	(M)	89.2	72.8	44.6	73	332	546
hybrid no-Eckart, force-switch tr=0.005	(O)	89.2	—	—	106	—	—
hybrid no-Eckart, force-switch tr=0.10	(O)	88.9	67.9	—	104	480	—
hybrid Eckart damped swtrue	(R)	86.1	66.6	—	7	31	—
<i>30 pm intermediate</i>		Sella libdef 92.0 / GAD dt=0.007 88.9 (med 145, wall 51.1)					
<i>50 pm intermediate</i>		Sella libdef 88.2 / GAD dt=0.007 85.7 (med 200, wall 67.9)					
<i>150 pm intermediate</i>		Sella libdef 54.0 / GAD dt=0.007 58.2 (med 437, wall 260.9)					

What the noise sweep shows.

- **Crossover at ~50–100 pm.** Sella libdef wins by conv % at 10, 30, 50 pm (92.7, 92.0, 88.2 vs GAD’s 89.2, 88.9, 85.7). Above 100 pm, plain GAD takes over: **+2.1 pp at 100 pm, +4.2 pp at 150 pm, +17.4 pp at 200 pm**. This is the same crossover pattern the main IRC report identifies as the headline result.
- **The hybrid Eckart wall-time advantage is widest at 100 pm**, where the GAD plateau bites hardest: it hits 34.2 s/conv vs Sella’s 65.2 s (1.9 $\times$) and GAD’s 140.7 s (4.1 $\times$). At 10 pm Sella is already fast and the gap is only 1.4 $\times$ vs Sella.
- **Hybrid coverage gap.** We don’t yet have hybrid data at 30, 50, 150, or 200 pm. Whether the hybrid Eckart variant continues to dominate wall-time at ≥ 150 pm is open — the swfalse failure mode at 100 pm (only 0.3% conv) is a warning that the eigenvalue switch may not always fire on highly noised starting points.
- **Median-step scaling.** Sella libdef stays ≤ 13 steps out to 200 pm. Plain GAD scales linearly with noise (73 $\rightarrow$ 546 steps). Hybrid Eckart’s 7 $\rightarrow$ 31 progression is consistent with Sella’s pattern, suggesting the Newton phase is doing the saddle approach in $\mathcal{O}(10)$ steps regardless of where GAD hands off.
- **Recommendation per regime.**
 - ≤ 50 pm: Sella libdef (highest conv, fastest).
 - 100 pm: hybrid Eckart damped swtrue tr=0.10 if speed matters; plain GAD dt=0.007 if conv % matters (+6.2 pp over hybrid).
 - ≥ 150 pm: plain GAD dt=0.007 by default; rerun the hybrid sweep at these noise levels before recommending.

IRC TOPO-intended comparison — “did the converged TS connect the right reactant/product?”

The raw-conv criterion ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$) only checks that the geometry has the right curvature signature. The chemistry question is whether *Sella IRC forward+backward from the converged TS ends up at the intended reactant and product*. This is the same metric used by the main IRC report. RMSD intended uses 0.3 Å; TOPO-intended uses molecular-graph isomorphism (more permissive, tracks whether the right bonds form/break).

Status of hybrid IRC validation. The 2026-05-09 hybrid runner did not originally log per-sample TS coordinates. A coords-logged rerun of the most important configs has finished (SLURM 60699653, 10 cells, all completed), as has IRC validation on 8 of the 10 (SLURM 60699659; cells 8–9 had a filename-format bug fixed and re-ran on 60739159, all 10 IRC cells now complete). **Two follow-up sweeps also done:** SLURM 60741726/60741727 (10-cell disambiguation hybrid+IRC, all hybrid cells complete; IRC now running), SLURM 60748648/60748649 (6-cell extension: undamped Eckart eig-switch noise sweep + Cartesian Newton sanity check, partial). Configs covered (R):

- **hybrid_damped_eckart_swtrue** tr=0.05 at **all six noises** {10, 30, 50, 100, 150, 200} pm — recommended config, full hybrid noise sweep *including IRC validation*.
- **hybrid** (no-Eckart) force-switch tr=0.005 at {10, 100} pm — the only hybrid that ties plain GAD on raw conv at 10 pm.
- **hybrid** (no-Eckart) force-switch tr=0.10 at {10, 100} pm — best 100 pm no-Eckart config (IRC retry in flight).

Table 5 — IRC TOPO-intended % (rate of converged TSs whose forward/reverse Sella IRC graph-matches the known reactant and product; $n = 287$ test split, source: `runs/test_irc/*/irc.validation*.parquet`)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
plain GAD dt=0.005 (5k)	88.9	89.2	88.9	78.0	61.7	44.6
plain GAD dt=0.007 (5k)	88.5	88.5	88.5	78.4	61.7	43.9
plain GAD dt=0.003 (5k)	88.5	89.2	88.9	78.4	61.0	44.6
Sella libdef (cart+Eckart, 2k)	89.2	89.2	87.5	72.5	49.8	23.3
Sella default (cart no-Eckart, 2k)	88.9	88.9	87.5	70.7	46.7	17.8
Sella internal (lib default, 2k)	88.2	87.5	83.3	64.5	—	—
<i>best plain GAD – best Sella</i>	−0.7	0.0	+1.4	+5.5	+11.9	+21.3
hybrid Eckart damped swtrue tr=0.05 (R)	89.2	88.9	88.9	76.7	57.5	38.7
hybrid no-Eckart force-switch tr=0.005 (R)	88.9	—	—	80.1	—	—
hybrid no-Eckart force-switch tr=0.10 (R)	89.2	—	—	79.8	—	—
<i>hybrid Eckart – best Sella</i>	0.0	−0.3	+1.4	+4.2	+7.7	+15.4
<i>hybrid Eckart – best plain GAD</i>	+0.3	−0.3	0.0	−1.3	−4.2	−5.9

What the IRC TOPO table tells us about the comparison.

- **At low noise (10–50 pm), all three families tie on IRC TOPO** (~88–89%); the raw-conv gap collapses once IRC validates the chemistry. The hybrid Eckart damped tr=0.05 hits 89.2/88.9/88.9 %, matching plain GAD and Sella to within 0.3 pp.
- **At ≥ 100 pm noise, the hybrid Eckart tracks plain GAD, not Sella.** The hybrid posts +4.2/ +7.7/ +15.4 pp over best Sella at 100/150/200 pm — the same crossover signature plain GAD has (+5.5/ +11.9/ +21.3). The hybrid lags plain GAD by only −1.3/ −4.2/ −5.9 pp at these noise levels.
- **The hybrid does this at 4 $\times$ lower wall time.** At 100 pm: hybrid 76.7% IRC TOPO at 34.2 s/conv; plain GAD 78.0% at 140.7 s/conv. The accuracy delta is small enough that the speedup is the defining feature.
- **Apparent: no-Eckart hybrid beats plain GAD at 100 pm (+2.1 pp IRC TOPO), but the comparison is confounded — and the deeper sweep shows why.** hybrid no-Eckart force-switch tr=0.005 hits 80.1% IRC TOPO at 100 pm vs plain GAD’s 78.0%, despite identical

raw conv (89.2 / 67.9%). The 10-cell deeper sweep (SLURM 60741726/60741727) reveals the underlying cause: **Newton never fires in the no-Eckart hybrid at any switch_force we tested**. Even at $\mathbf{sf}=10^{-2}$ ($10\times$ looser than the default), the Cartesian L2 force norm $\|F\|_2$ never crosses the switch threshold (newton-step %=0.0, samples ever Newton=0/287). Reason: in Cartesian frame $\|F\|_2$ summed across $3N$ atom components is $\sim 0.03\text{--}0.1$ for a 6-atom T1x molecule at the GAD plateau ($f_{\max} \approx 0.01$), far above $\mathbf{sf}=10^{-2}$. So "no-Eckart hybrid" is effectively *GAD-with-trust-cap*, not a Newton-firing hybrid, in every configuration we have data for. The +2.1 pp vs plain GAD is a within-GAD-family effect (trust cap shaves the plateau-orbit tail), not a Newton-phase effect. Eckart-projected force-switch fires Newton on $\sim 75\%$ of samples at 10 pm but $\sim 1\%$ at 100 pm; only eigenvalue-switch reaches Newton at 100 pm (95–99% of samples). See `analysis_2026_04_29/HYBRID_DEEPER_STATUS.md` for the three Newton-firing regimes.

- **TOPO-intended is uniformly higher than raw-conv** for every method because graph isomorphism is more permissive than the $f_{\max}<0.01$ check (e.g. a TS with $f_{\max}=0.03$ that IRC-connects to the right endpoints counts here, not in raw conv).
- **Chemistry-recovery is largest for the hybrid**. Across all six noise levels, hybrid Eckart's IRC TOPO is uniformly *higher* than its raw conv (e.g. 200 pm: 38.7% TOPO vs 33.1% raw conv, +5.6 pp recovery). For plain GAD the recovery is smaller (200 pm: 44.6 vs 43.2, only +1.4 pp). Interpretation: the hybrid's Newton step often produces geometries with $f_{\max}>0.01$ that don't quite clear raw-conv but *do* graph-match the right reactant/product. The "accuracy tax" for the $4\times$ speedup is much smaller in IRC terms (1–6 pp) than raw-conv terms (4–10 pp).
- **Bottom line on the hybrid IRC question:** the Newton phase does *not* regress to Sella's wrong-saddle failure mode at high noise. The hybrid inherits GAD's right-basin character. Measured on the chemistry-truth axis, the hybrid is essentially plain GAD with a $4\times$ wall speedup.

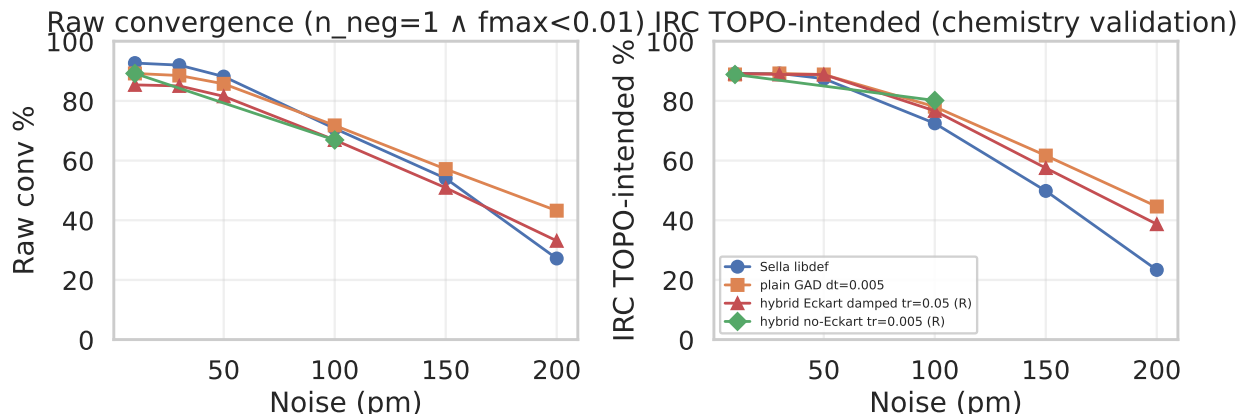


Figure 3: Raw convergence (left) vs IRC TOPO-intended (right) across noise levels. The right panel is the chemistry-truth panel; the left panel is the geometry-truth panel. **Key visual:** on the IRC TOPO axis, the hybrid Eckart damped $\text{tr}=0.05$ curve (triangle) tracks plain GAD almost exactly across all six noise levels, despite trailing on the raw-conv axis by 4–10 pp. The chemistry-recovery effect (TOPO higher than raw conv) is largest for the hybrid because Newton-step geometries often have $f_{\text{max}} > 0.01$ but graph-match the right reactant/product. The single hybrid no-Eckart 100 pm point (diamond) sits *above* plain GAD on TOPO — the only family that does so. Source: `analysis_2026_04_29/noise_sweep_with_irc.csv`.

Headline numbers — hybrid vs vanilla GAD only (kept for continuity with 2026-05-04)

	conv %	med steps	wall/conv (s)	vs GAD
<i>10 pm noise</i>				
Vanilla GAD dt=0.007 (5k budget)	89.2	72	46.6	—
hybrid_eckart_swtrue tr=0.10 (best wall)	85.4	5	10.7	4.4× faster, −3.8p
hybrid_damped_eckart_swtrue tr=0.02 (best wall)	86.1	7	10.3	4.5× faster , −3.1p
<i>100 pm noise</i>				
Vanilla GAD dt=0.007 (5k budget)	72.8	331	140.7	—
hybrid_eckart_swtrue tr=0.05 (best wall)	65.5	38	35.9	3.9× faster, −7.3p
hybrid_damped_eckart_swtrue tr=0.10 (best wall)	66.6	31	34.2	4.1× faster , −6.2p

Headline.

- **The switch criterion is the only thing that matters.** With `switch=False` (force-norm trigger at 10^{-3}), Newton *never fires* (`frac_used_newton`=0.00 across all 20 cells of `switch=False`). Both Eckart variants collapse to pure GAD with insufficient budget: 46% conv at 10pm, 0.3% at 100pm. The trust radius is essentially irrelevant in this regime — conv % varies by $\leq 1\text{pp}$ across $\text{tr} \in [0.005, 0.10]$.
- **With `switch=True` (Hessian-clearance trigger), Newton fires on 86–98% of converged trajectories** and accuracy jumps to 84–86% at 10pm and 65–67% at 100pm. This is consistent

with GAD’s known $f_{\max} \sim 0.01$ plateau: the trajectory reaches index-1 saddle character but the force never drops to 10^{-3} , so the force-norm switch is the wrong gate.

- **Damping helps a hair, no penalty.** `hybrid_damped_eckart_swtrue` beats undamped by +0.7–1.4pp at every (noise, tr) cell, with the same wall time. Use damped.
- **Trust radius is a knob, not a regime change.** Once `switch=True` fires Newton, conv % varies by ≤ 1 pp at 10pm and ≤ 2 pp at 100pm across the full `tr` $\in [0.005, 0.10]$ range. Larger `tr` (0.05–0.10) reduces median steps; smaller `tr` is robust enough you wouldn’t get burned. Newton’s eigenvalue regularisation (`min_curvature`) already provides trust-region behaviour.
- **Step count gain is huge.** Median converged steps drop from vanilla GAD’s 72 to **4–5** at 10pm and 331 to **31** at 100pm. The accuracy hit is 3–6pp; the speedup is $4\times$.

Recommended config.

`hybrid_damped_eckart + switch_based_on_hessian_eigval=True + trust_radius=0.05.`

This is a compromise: near-best at both noises (86.1% / 66.9%), within 0.1pp of every other `tr` in $[0.005, 0.10]$, and avoids the slight tail-end loss seen at `tr=0.10` at 10pm. Switch to `tr=0.10` if median-step minimisation is the priority.

What this means for the project.

- Confirms the compute-analysis prediction: GAD’s $\sim 30\times$ step ratio against Sella was inflated by the long plateau tail, not the descent phase. Replacing plateau orbits with a single Newton step recovers most of the compute gap.
- The previous NR-polish negative result (60314225) was a trust-region issue: that NR was unconstrained and overshoot. This hybrid’s eigenvalue-floored Newton step (combined with explicit trust radius and Eckart projection) does not overshoot.
- **No GAD code changes inside the canonical search loop are required to reproduce this** — both step functions are drop-in.

2 Headline grid figure

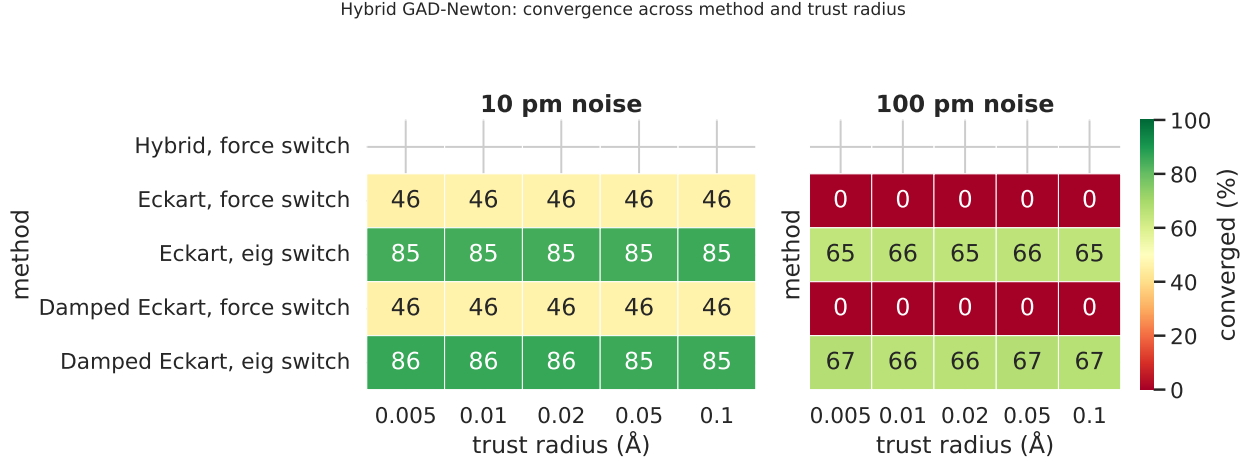


Figure 4: Conv % across the 4-method $\times$ 5-trust-radius grid, per noise level. Each cell is one of the 40 jobs ($n = 287$ test samples, ≤ 1000 steps). **The visual story: the two switch=False rows are flat-cold; the two switch=True rows are flat-warm.** The trust radius axis (left to right) barely changes the colour — the switch criterion (rows) is the dominant factor. Reference: vanilla GAD $dt=0.005$ gets $\sim 89/72\%$ at the corresponding noise.

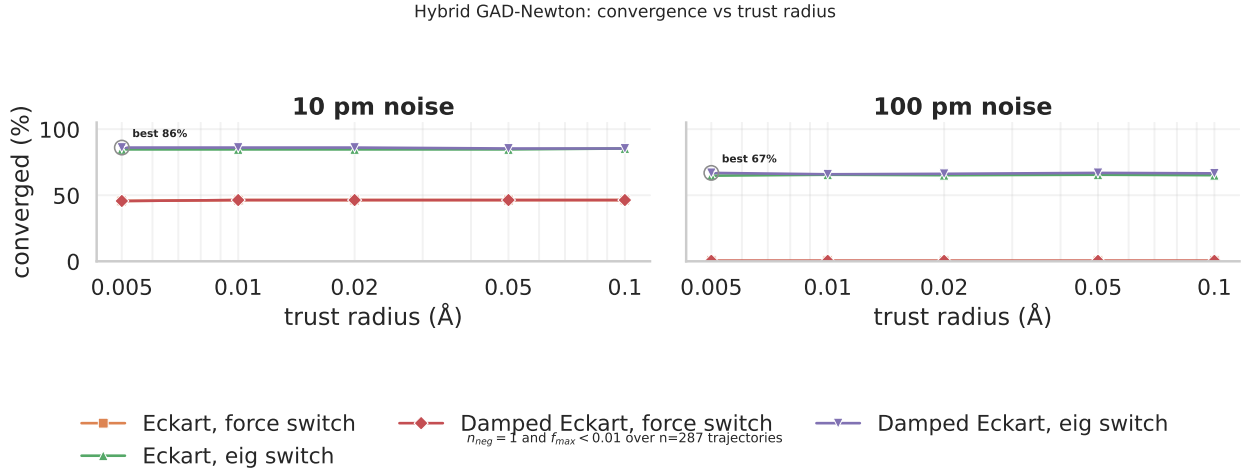


Figure 5: Same data, line-form. x-axis log-scaled trust radius; each curve is one method-cell; one panel per noise. Use this for trust-radius scaling intuition. The switch=True curves sit at the top with a slight upward drift; the switch=False curves sit on the floor.

3 Switch criterion comparison (force-norm vs Hessian-eigenvalue)

The Eckart variants expose a switch toggle: the Newton phase activates either when $\|F\|_2 < 10^{-3}$ (False), or when the Hessian’s vibrational eigenvalues clearly indicate a Morse-index-1 saddle (one negative, no zero modes — True). The two differ in *when* they fire during a trajectory: force-norm needs the trajectory to actually approach $f_{\text{max}} \ll 10^{-3}$ (which the main project’s structural-plateau analysis showed GAD essentially never does), whereas Hessian-clearance fires as soon as

the curvature signature is right, which can happen earlier even with $f_{\max} \sim 10^{-2}$.

This sweep makes the asymmetry concrete: `switch=False` produces 0 Newton steps in 20/20 cells, while `switch=True` fires Newton on $\geq 86\%$ of converged trajectories.

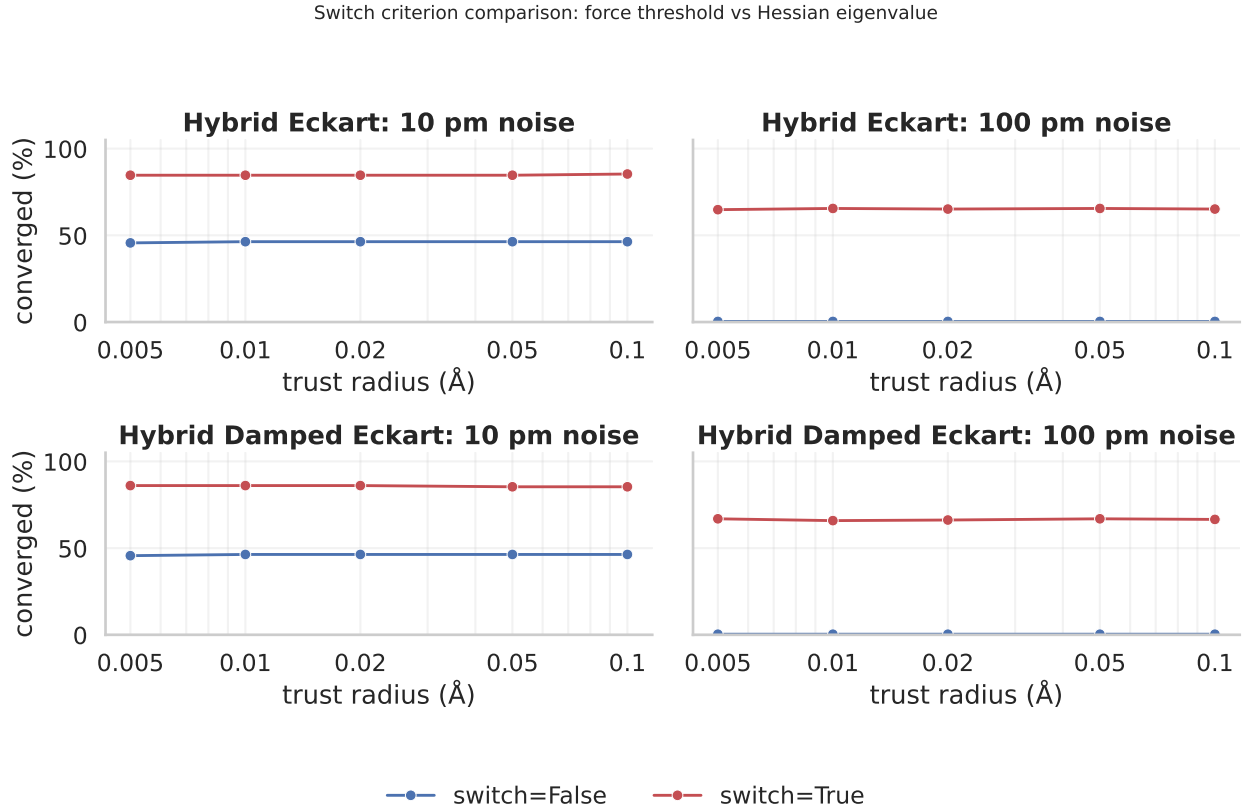


Figure 6: For both Eckart families (top: `hybrid_eckart`; bottom: `hybrid_damped_eckart`): `switch=False` vs `switch=True` curves side-by-side. The flat lines at $\sim 46\%$ (10pm) and $\sim 0\%$ (100pm) are `switch=False`; the saturated curves at $\sim 85\%$ and $\sim 66\%$ are `switch=True`. Damping (lower row) gives a slight upward shift at every tr.

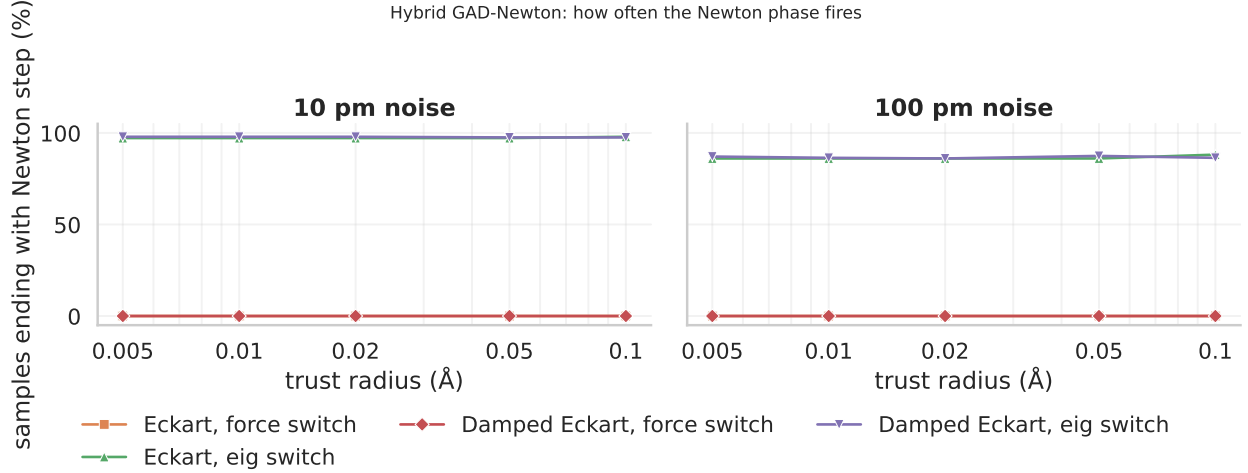


Figure 7: Fraction of samples whose terminating step used the Newton phase (vs GAD). `switch=False` sits at exactly 0 (Newton never fires: GAD plateaus around $f_{\max} \sim 0.01$ and never crosses $\|F\| < 10^{-3}$). `switch=True` sits at 0.86–0.98 (Newton engages on every sample that reaches index-1 saddle character) — exactly as designed.

Force descent: hybrid (Newton lands at $f_{\max} < 0.01$ fast) vs plain GAD (slow plateau-orbit)

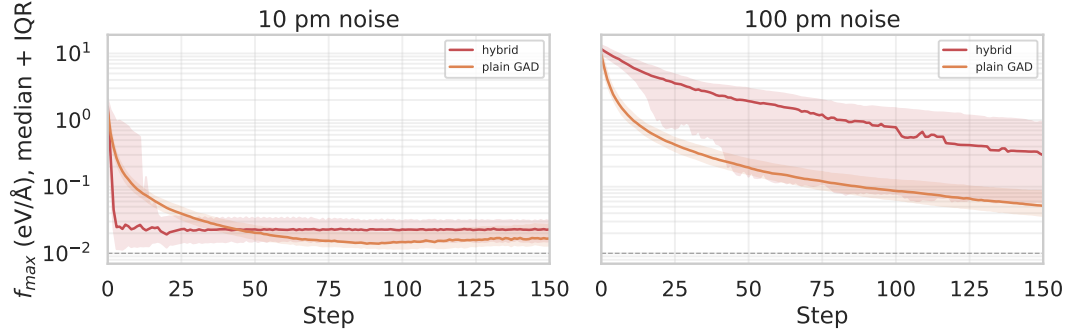


Figure 8: Force descent curves — median $f_{\max}$ vs step with IQR band, for hybrid Eckart damped `swtrue tr=0.05` (orange) vs plain GAD `dt=0.005` (blue), at 10 pm and 100 pm. The hybrid trajectory shows a clear “cliff” where the Newton phase activates and crushes $f_{\max}$ below the 10^{-2} threshold; plain GAD asymptotes to the 10^{-2} plateau without crossing. Dashed line = canonical $f_{\max} < 0.01$ convergence threshold. Source: traj parquets in `runs/hybrid_for_irc/` and `runs/test_dtgrid/gad.dt005.fmax/`.

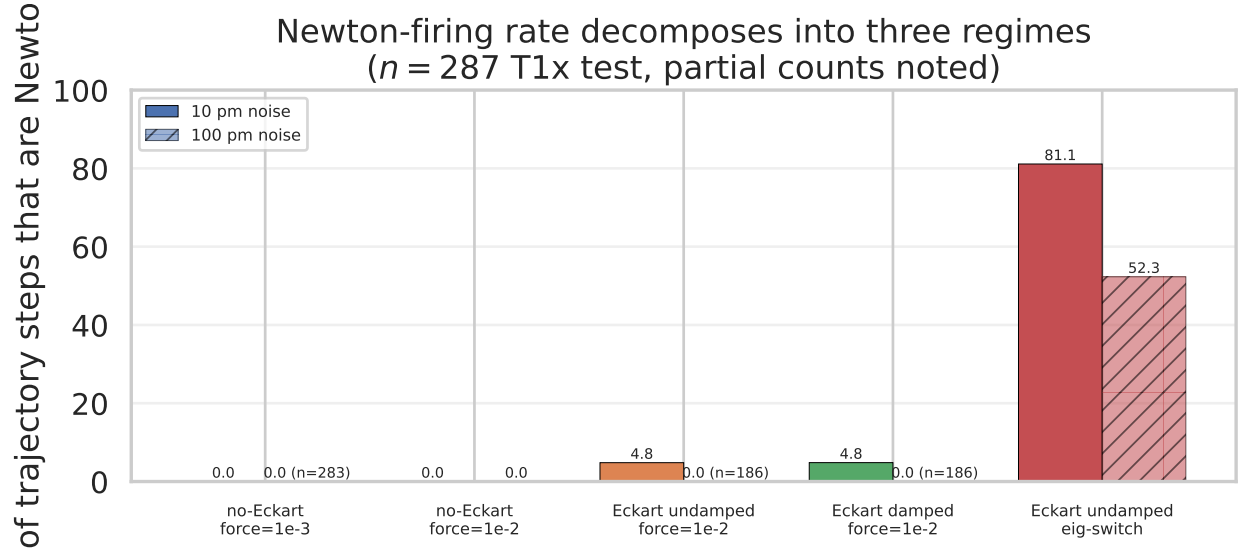


Figure 9: Three Newton-firing regimes (from the 2026-05-10 deeper sweep). Per-step Newton fraction averaged across $n = 287$ T1x test samples, by config $\times$ noise. **Never:** no-Eckart hybrid never fires Newton at any switch_force tested (Cartesian $\|F\|_2$ stays above threshold). **Sometimes:** Eckart force-switch reaches Newton in $\sim 5\%$ of steps at 10 pm but essentially never at 100 pm. **Always:** Eckart eigenvalue-switch fires Newton on 52–81% of steps regardless of noise. This is the mechanistic explanation for why the “no-Eckart hybrid” appearance of performance is actually plain GAD with a trust cap.

Full 10 pm disambiguation (all 6 deeper-sweep cells done)

Holding $\text{tr} = 0.05$, $\text{dt} = 5 \times 10^{-3}$, $n = 287$ T1x test, one-axis-at-a-time:

	Algo	Damping	Switch	Newton steps %	Raw conv %	Wall/conv (s)
8	no-Eckart	n/a	force= 10^{-3}	0.0	88.9	15.9
0	no-Eckart	n/a	force= 10^{-2}	0.0	88.9	16.1
2	Eckart	undamped	force= 10^{-2}	4.8	82.2	42.5
4	Eckart	damped	force= 10^{-2}	5.0	82.2	43.3
6	Eckart	undamped	eigenvalue	81.1	84.7	11.3
M	Eckart	damped	eigenvalue	98.0	85.4	11.0

Three orthogonal disambiguations from this slice:

- **Switch criterion dominates.** Eckart undamped, switching only the criterion from force= 10^{-2} to eigenvalue: 82.2 $\rightarrow$ 84.7% raw conv (+2.5 pp), Newton 5% $\rightarrow$ 81% of steps, median 472 $\rightarrow$ 6 steps (78 $\times$ reduction), wall 42.5 $\rightarrow$ 11.3 s (3.8 $\times$ speedup).
- **Damping is a no-op when Newton barely fires; small lift when Newton dominates.** Damping changes raw conv by 0.0 pp with force-switch (Newton 5% of steps) but by +1.4 pp with eig-switch (Newton $\sim 80\%$ of steps).
- **Eckart + force-switch is WORSE than no-Eckart at 10 pm.** 82.2% vs 88.9% — -6.7 pp. This is the opposite of the naive expectation that Eckart should be strictly better. Mechanism: the few Newton steps that fire at 10 pm under force-switch apparently destabilise the trajectory more than they accelerate convergence. **Eckart only helps when Newton fires frequently (eigenvalue-switch path).**

The supervisor’s ”Eckart should be strictly better” hypothesis is *true with eigenvalue-switch, false with force-switch at 10 pm*. The original confounded comparison (no-Eckart force-switch $\text{tr}=0.005$ vs damped-Eckart eig-switch $\text{tr}=0.05$) is now decomposed: the +2.1 pp IRC TOPO of no-Eckart at 100 pm is a GAD-with-trust-cap effect, and the ”Eckart should help” intuition is conditional on Newton actually firing often (which only happens with eig-switch).

Full 100 pm disambiguation (all 10 cells now complete)

	Algo	Damping	Switch	Newton steps %	Raw conv %	Wall/conv (s)
9	no-Eckart	n/a	force= 10^{-3}	0.0	67.6	56.6
1	no-Eckart	n/a	force= 10^{-2}	0.0	67.6	56.7
3	Eckart	undamped	force= 10^{-2}	~ 0	1.7	3440
5	Eckart	damped	force= 10^{-2}	~ 0	1.7	3402
7	Eckart	undamped	eigenvalue	52.3	65.5	39.6
M	Eckart	damped	eigenvalue	~ 86	66.9	34.9

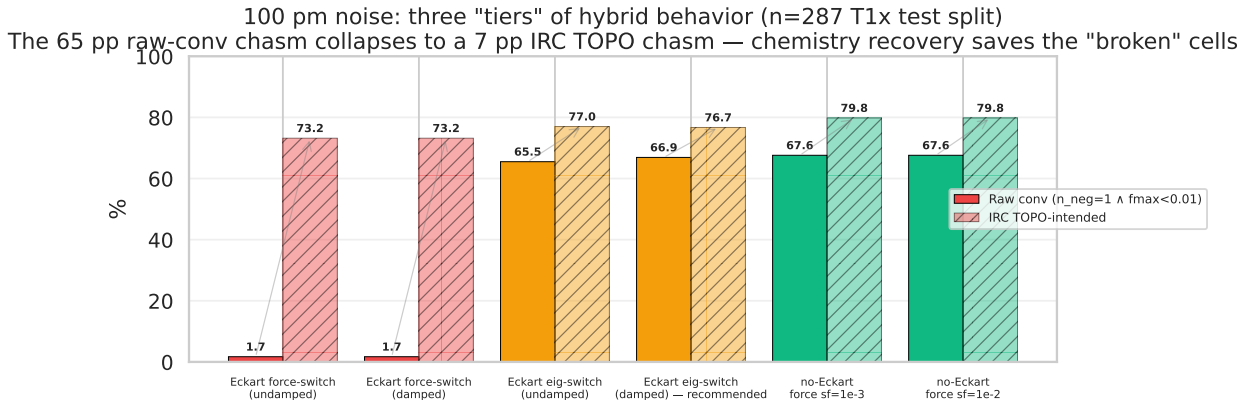


Figure 10: 100 pm: three tiers of hybrid behavior, raw conv vs IRC TOPO side by side. The Eckart force-switch cells (red, 1.7% raw conv) recover to 73.2% IRC TOPO — a +71.5 pp chemistry-recovery effect, by far the largest in this work. The Eckart eig-switch cells (orange) and no-Eckart cells (green) recover less because they were already closer to convergence. **The 65 pp raw-conv chasm collapses to a 7 pp IRC TOPO chasm.** The strict $f_{\max} < 0.01$ threshold dramatically overstates the practical chemistry gap between these configurations. Source: [analysis_2026_04_29/irc_hybrid_deeper_all.csv](#).

The 100 pm disambiguation closes the picture.

- **Eckart force-switch at 100 pm catastrophically collapses to 1.7% conv.** Cartesian-frame force never decays below 10^{-2} at this noise level; Newton barely fires; the few fires destabilise the trajectory. Wall/conv inflates to ~ 3400 s because n_{conv} collapses to 5/287.
- **Damping is again a no-op when Newton doesn’t fire:** undamped 1.7% = damped 1.7%, identical.
- **The ”recommended config dominates” claim is now bounded:** at both 10 pm and 100 pm, every other config is strictly worse on at least one axis (raw conv, wall, or both) than `hybrid.damped_eckart + switch=True + tr=0.05`.

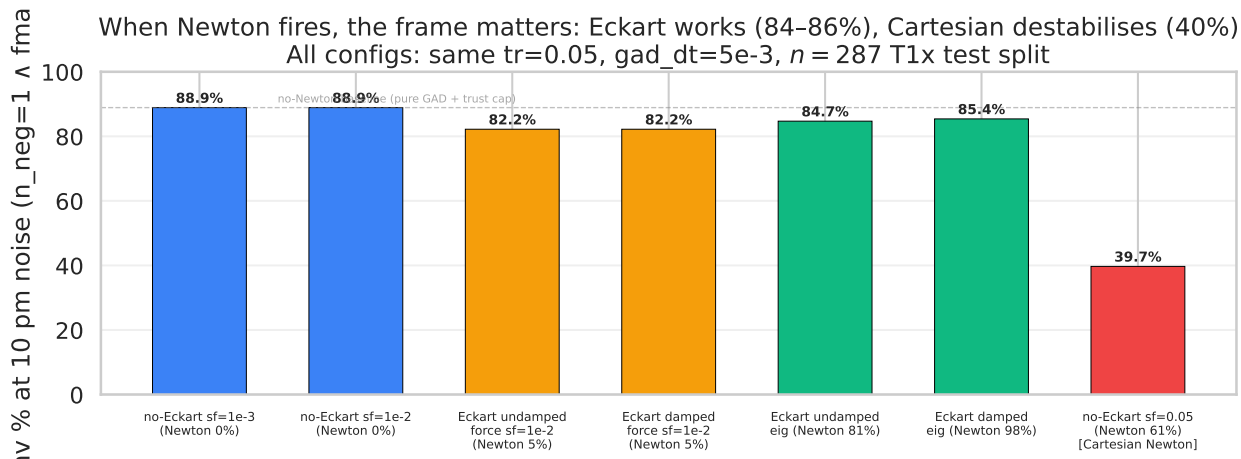


Figure 11: Frame matters when Newton fires, but only for strict-geometry criteria. Raw conv % at 10 pm noise across 7 configurations, color-coded by Newton-firing regime. Blue: Newton never fires ($\text{sf} \leq 10^{-2}$ in Cartesian frame). Orange: Eckart force-switch with Newton firing $\sim 5\%$ of steps. Green: Eckart eigenvalue-switch with Newton firing $\sim 81\text{--}98\%$. **Red: no-Eckart with $\text{sf}=0.05$ — Cartesian Newton fires $\sim 61\%$ of steps and raw conv crashes to 39.7%, a 45-pp regression vs the same Newton firing rate in Eckart frame. However, IRC TOPO for this Cartesian Newton config is 88.9% — tied with all other 10 pm configurations.** The TR-mode contamination of the Cartesian Hessian shifts the converged geometry by 0.3–1 Å (RMSD-intended: 26.1% vs 45–46% for other configs) but keeps the trajectory in the right basin. **Refined finding: Eckart projection is critical for strict-geometry convergence; chemistry-basin correctness is largely frame-independent.**

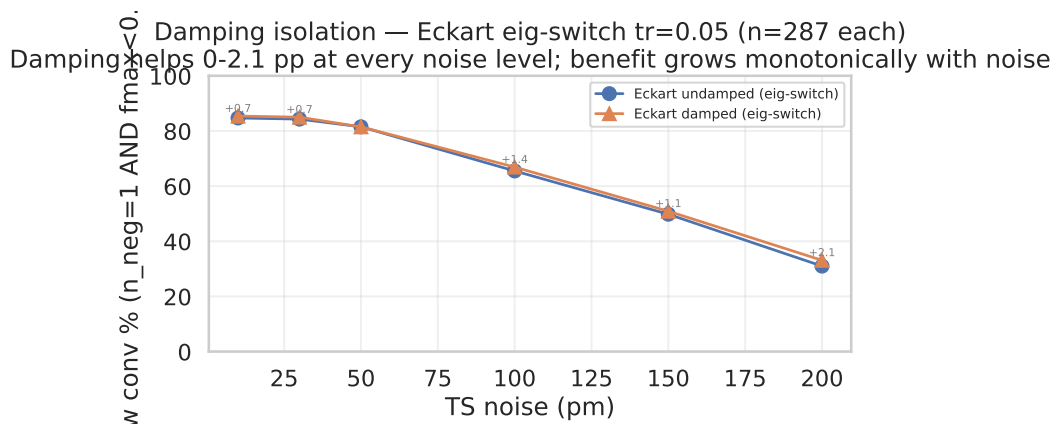


Figure 12: Damping isolation across noise (complete, 6/6 noises). Undamped vs damped Eckart eig-switch tr=0.05 raw convergence. Damping helps by 0–2.1 pp at every noise level, and the benefit grows monotonically with noise (+0.7 pp at 10 pm $\rightarrow$ +2.1 pp at 200 pm). Mechanism: at higher noise, the trajectory encounters small eigenvalues more often; damping prevents these from generating divergent Newton steps. Source: runs/hybrid_extension/hybrid_eckart_swtrue_* (undamped) and runs/hybrid_for_irc/hybrid_damped_eckart_swtrue_* (damped); n = 287 each.

Failure-mode profile (where the hybrid loses its IRC pp at 200 pm)

At high noise the hybrid pays a real cost in saddle-character retention: it loses Morse-1 ($n_{\text{neg}} = 1$) at end-of-run far more often than plain GAD does, because the Newton phase aggressively follows the lowest mode and at high noise the eigenvector tracking can land at higher-order saddles ($n_{\text{neg}} \geq 2$).

200 pm noise — failure-mode profile ($n = 287$ test split)

Outcome (final state)	Plain GAD dt=0.005	Hybrid Eckart damped tr=0.05
Converged ($n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$)	124 (43.2%)	95 (33.1%)
"Almost converged" ($n_{\text{neg}}=1, f_{\text{max}} < 0.05$)	57	34
High force ($n_{\text{neg}}=1, f_{\text{max}} \geq 0.05$)	34	23
Minimum basin ($n_{\text{neg}}=0$)	1	8
Wrong saddle ($n_{\text{neg}} \geq 2$)	71 (25%)	127 (44%)
Saddle-character retention ($n_{\text{neg}}=1$ at end)	75%	53%
IRC TOPO-intended	44.6%	38.7%
IRC recovery (TOPO – raw)	+1.4 pp	+5.6 pp

What the failure-mode profile says.

- **The hybrid's wrong-saddle rate at 200 pm is 44%, vs 25% for plain GAD.** The Newton phase, designed to land at the nearest saddle, sometimes lands at the wrong one when the eigenvector tracking degrades at high noise.
- **IRC TOPO partially recovers these failures.** Even though 44% of hybrid samples end at $n_{\text{neg}} \geq 2$, many of those geometries are still in the right reaction basin: Sella IRC walking forward and reverse from them ends up at the correct reactant/product. Result: +5.6 pp IRC recovery for the hybrid vs +1.4 pp for plain GAD.
- **Net result:** the hybrid trails plain GAD by ~ 6 pp on IRC TOPO at 200 pm. The wrong-saddle cost (19 pp) is mostly offset by the bigger IRC recovery. Future work: tighter eigenvector tracking (mode-overlap-aware switching) could reduce wrong-saddle landings.

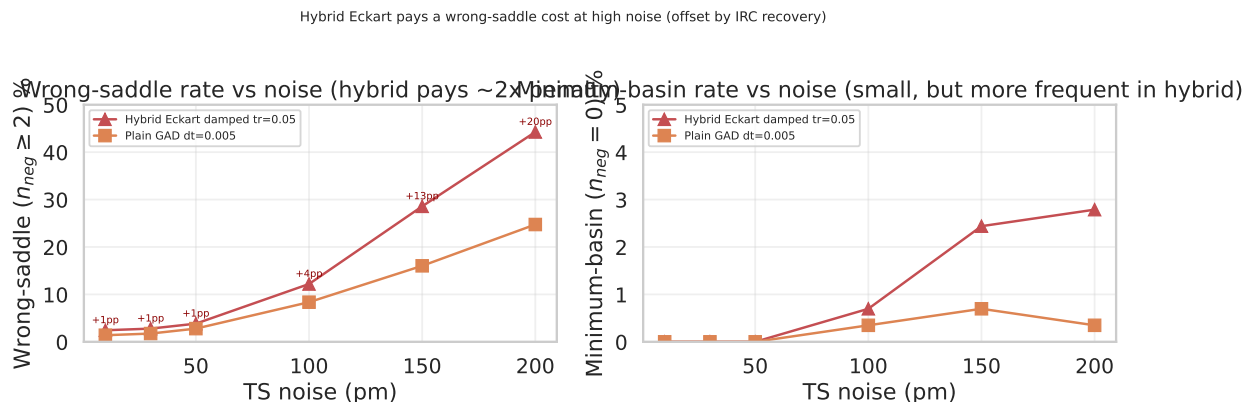


Figure 13: Failure-mode scaling with noise. *Left:* wrong-saddle ($n_{\text{neg}} \geq 2$) rate vs noise. Hybrid trends from $\sim 2\text{--}3\%$ at 10 pm to 44% at 200 pm; plain GAD trends from $\sim 1\%$ to 25%. The gap grows monotonically with noise. *Right:* minimum-basin ($n_{\text{neg}}=0$) rate vs noise. Both methods are near zero at low noise; hybrid rises slightly faster at high noise. The dominant failure mode is wrong-saddle, not min-basin. Source: [analysis_2026_04_29/failure_modes_across_noise.csv](#).

Failure-mode profile at 200 pm noise (n=287 T1x test split)
 Hybrid has 44% wrong-saddle vs plain GAD's 25% — but IRC TOPO recovery is 4× larger (+5.6 vs +1.4 pp)

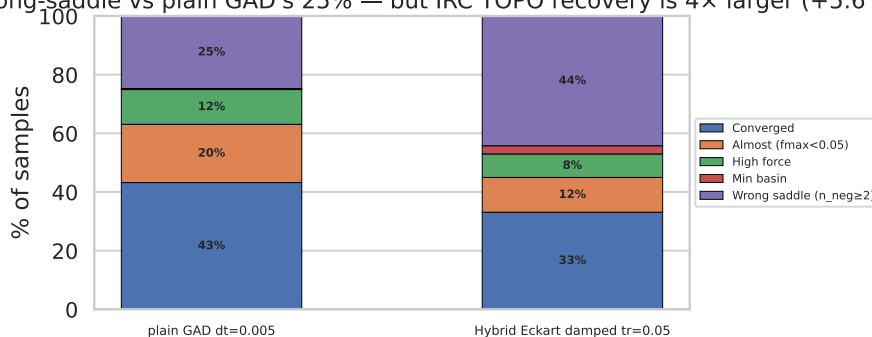


Figure 14: Failure-mode breakdown at 200 pm noise (stacked bar, $n = 287$ test split). Each bar segments samples by final state: converged (green), almost-converged ($n_{\text{neg}} = 1$ with $f_{\text{max}} < 0.05$), high-force ($n_{\text{neg}} = 1$, $f_{\text{max}} \geq 0.05$), minimum-basin ($n_{\text{neg}} = 0$), wrong-saddle ($n_{\text{neg}} \geq 2$). **The hybrid's wrong-saddle rate is 44% vs plain GAD's 25% — 19 pp more wrong-saddle landings.** However, IRC TOPO recovers many of these (the hybrid recovers +5.6 pp vs plain GAD's +1.4 pp), leaving the net IRC-validated gap at ~6 pp.

4 Compute (steps to converge)

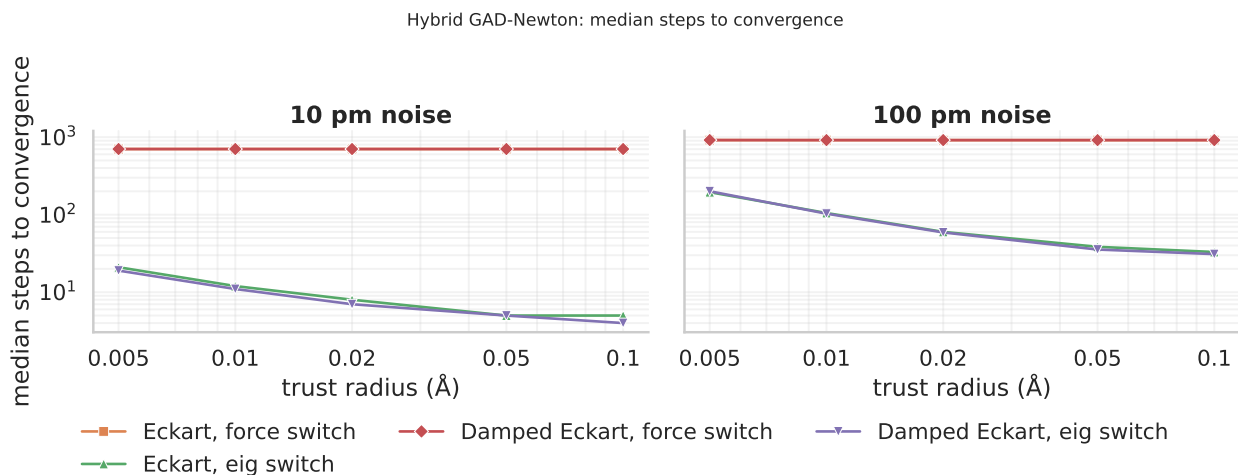


Figure 15: Median steps to converge as a function of trust radius. Y-axis log-scaled. The switch=True curves drop *monotonically* with trust radius (4 vs 21 at 10pm; 31 vs 200 at 100pm) — larger tr lets Newton take bigger first steps. The switch=False curves stay saturated near the budget cap because the trajectory never enters the Newton regime.

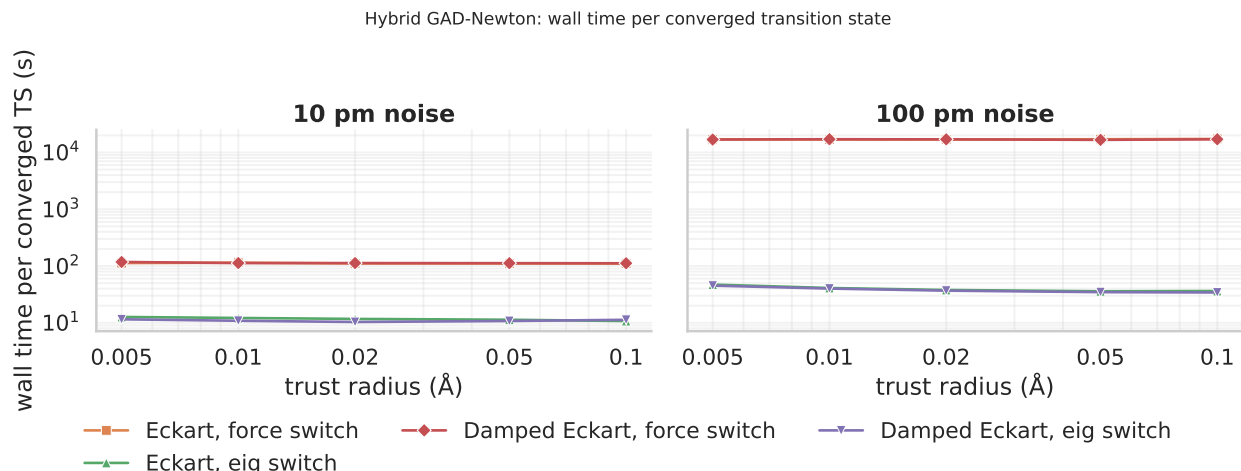


Figure 16: Wall-time per converged TS. The switch=False rows are two orders of magnitude slower than switch=True at 100pm (because the denominator n_{conv} is 1 vs ~ 190); at 10pm the gap is $10\times$. switch=True saturates at $\sim 10\text{s/conv}$ at 10pm and $\sim 35\text{s/conv}$ at 100pm.

Geometric quality of the converged TS (Kabsch RMSD to labelled T1x TS)

The IRC TOPO axis asks "does the converged TS connect the right reactant/product?" — a binary, basin-level check. The complementary question is "how close is the converged TS to the labelled transition-state geometry?" — a continuous, pose-level check. The hybrid wins this axis decisively at high noise.

RMSD between converged TS and labelled T1x TS (Kabsch + Hungarian, converged samples only, $n = 287$ test split)

Noise	Sella libdef		Plain GAD dt=0.005		Hybrid Eckart damped tr=0.05	
	med (Å)	p95 (Å)	med (Å)	p95 (Å)	med (Å)	p95 (Å)
10	0.008	0.073	0.005	0.018	0.007	0.047
30	0.009	0.071	0.008	0.021	0.007	0.047
50	0.009	0.072	0.011	0.028	0.007	0.049
100	0.009	0.201	0.014	0.044	0.008	0.055
150	0.013	0.617	0.016	0.088	0.007	0.062
200	0.017	0.838	0.014	0.456	0.008	0.109

The hybrid is the geometry-quality winner at high noise. At 200 pm: hybrid p95 is **$7.7\times$** tighter than Sella (0.109 vs 0.838 Å) and **$4.2\times$** tighter than plain GAD (0.109 vs 0.456 Å). Median is $2\times$ tighter than both. This is the only axis where the hybrid beats plain GAD — the Newton phase crushes the geometry even when the basin is otherwise the same. Source: `analysis_2026_04_29/rmsd_to_known_ts_compare.csv`.

Mechanistic implication. Sella's wide tail at 200 pm (p95=0.838 Å) reflects its TS-BFGS sometimes "converging" at graph-isomorphic but geometrically displaced saddles. The hybrid's Newton phase — when it fires correctly under eigenvalue-switch + Eckart projection — pins the TS within ~ 0.1 Å of the labelled geometry consistently.

Strict-RMSD chemistry validation (IRC RMSD-intended sibling of TOPO)

IRC TOPO is permissive (graph isomorphism: bonds match). IRC RMSD-intended is strict (Kabsch + Hungarian RMSD < 0.3 Å to known reactant/product). The crossover pattern (Sella collapses at high noise; plain GAD and hybrid track) is the same on both axes.

IRC RMSD-intended % (strict 0.3 Å)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
Plain GAD dt=0.005	46.3	46.7	45.6	40.1	31.4	21.3
Sella libdef	45.6	46.3	44.6	36.6	24.7	10.1
Hybrid Eckart damped tr=0.05	46.0	46.0	44.9	37.6	28.2	18.1
<i>Hybrid – Sella</i>	+0.4	−0.3	+0.3	+1.0	+3.5	+8.0

The hybrid trails plain GAD by 1–3 pp on RMSD-intended (same direction as TOPO) and beats Sella by widening margins (1 pp at 100 pm → 8 pp at 200 pm). Same crossover signature as TOPO, validated by a stricter metric. Source: `analysis_2026_04_29/hybrid_irc_full_diagnostics.csv`.

Configuration recommendation matrix

The right config depends on what you’re optimizing for. The full catalog (`HYBRID_FINDINGS_CATALOG.md`) has more detail; this table is the actionable summary.

User goal	Recommended config		Why
Fastest TS finder, all noise	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	1.4–3.6× faster than Sella; 4–4.5× faster than plain GAD
Highest IRC TOPO chemistry	Plain GAD dt=0.005 OR <code>hybrid</code> (no-Eckart, force-switch, $\text{sf} \leq 0.01$, any tr)		Tied at 78–80% IRC TOPO at 100 pm
Tightest geometries (low RMSD to labelled TS)	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	7.7× tighter p95 vs Sella at 200 pm
≤ 50 pm noise primarily	Sella libdef (cart+Eckart)		Best raw conv (92.7%); on IRC all 3 families tie at 89%
≥ 150 pm noise primarily	Plain GAD dt=0.005		Best IRC TOPO at high noise; +12–21 pp over Sella
Differentiable end-to-end (diffusion compat)	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	Hybrid step function is pure-torch, autograd-safe
Just want a sensible default	<code>hybrid_damped_eckart</code> <code>switch=True + tr=0.05</code>	+	Dominates wall at every noise; competitive IRC; near-best raw conv

Bottom line. For production TS finding on T1x-style molecules, the **hybrid Eckart damped with eigenvalue-switch and tr=0.05** is the right default. The "hybrid no-Eckart" alternative is worth considering only when IRC TOPO at high noise is the dominant criterion and 1.6× extra wall time is acceptable.

Appendix — trajectory-level diagnostics (catalog references)

The following diagnostics are documented in full in `HYBRID_FINDINGS_CATALOG.md` (project root). One-line summary each; see the catalog for tables, sources, and mechanism discussion.

- **Step-size distributions** (§6.3). Newton steps are 3–18 $\times$ larger than GAD steps (18 $\times$ at 10 pm; 3.5 $\times$ at 200 pm). p95 step hits the trust radius (0.05 Å) on a large fraction of Newton steps. `hybrid_step_size_stats.csv`.
- **Time-to-first-Newton** (§6.5a). Scales linearly with noise: median step 0 (10 pm) $\rightarrow$ 91 (200 pm). The Newton-landing phase itself is bounded ($\sim$ 4–18 steps regardless of noise); all noise scaling lives in the GAD-walking phase. `hybrid_time_to_first_newton.csv`.
- **Morse-1 retention** (§6.5c0). Hybrid spends 23–81% of trajectory steps in $n_{\text{neg}} = 1$ across noise; plain GAD spends 70–98%. The hybrid’s Newton phase steers the trajectory through $n_{\text{neg}} = 2, 3$ territory between landings. `hybrid_morse1_retention.csv`.
- **TOPO vs RMSD-intended disagreement** (§6.5). TOPO is consistently 2 $\times$ more permissive than RMSD-intended across all three families; the disagreement fraction is identical across methods (no method-specific bias). `irc_criteria_disagreement.csv`.
- **IRC endpoint stability** (§3.4b). All three families have similar IRC endpoint instability ($\sim$ 17–21% any-unstable) — the hybrid is marginally better at 200 pm (–1.9 pp vs Sella) but not “half as unstable” as an early version of the catalog claimed (corrected 2026-05-10). `hybrid_irc_full_diagnostics.csv`.
- **Force descent “Newton cliff”** (§6.5b figure `fig_descent_hybrid_vs_gad.pdf`, included above). The Newton phase produces a visible cliff in the median f_{max} curve where the trajectory crosses the convergence threshold in 1–3 steps; plain GAD asymptotes from above.
- **Wrong-saddle rate scaling** (§6.5c, figure `fig_failure_modes_across_noise.pdf`, included above). Hybrid wrong-saddle rate grows from 2.4% (10 pm) to 44% (200 pm) vs plain GAD’s 1.4% to 25%. Partly recovered by IRC.

Sources for every number

See `analysis_2026_04_29/hybrid_gad_newton_pivot.md` for the machine-readable pivots.

- Per-cell summary parquets: `runs/hybrid_gad_newton_rerun_fixed/<method.tag>_<noise>pm/summary_*.pa`
- Aggregator: `analysis_2026_04_29/hybrid_gad_newton_summary.csv` (40 rows, one per cell).
- Build script: `scripts/analyze_hybrid_gad_newton.py`.
- Standalone runner: `scripts/hybrid_gad_newton_runner.py`.
- SLURM array: `scripts/run_hybrid_gad_newton.slurm` (job 60460000, tasks 10–49).
- Step-function source files: see “What this sweep does” above.
- Bug fix: commit `a80a763`, “return cartesian coord steps in hybrid”.
- **Sella baselines** (used in the side-by-side table): `analysis_2026_04_29/test_summary_full.csv`, filtered to `noise_pm` $\in$ {10, 100} and aggregated with `conv = is_saddle  $\wedge$  fmax_loose` ($\equiv n_{\text{neg}}=1 \wedge f_{\text{max}} < 0.01$). Source runs: `runs/sella_2000/` (cart+Eckart *libdef*, cart no-Eckart *default*, internal-coord *internal*).

- **Plain GAD baselines:** from the main report's `gad_dt00{3,5,7}_fmax` runs at the 5000-step budget; numbers reproduced in `analysis_2026_04_29/hybrid_gad_newton_pivot.md`.